

BIOL343 Coding Challenge 3

Group Members

2026

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Group Members

```
# Name - Queen's Student Number - Github repo  
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```

Part 1 — Import, Inspect, and Summarize

1. Import the Dataset

Import the provided dataset using `read.csv()` and save it as `floral_data`.

```
floral_data <- read.csv("data/AbroniaFloralData.csv")
```

2. Inspect the Dataset

Use the following functions:

- `dim()`
- `head()`
- `tail()`
- `str()`

```
# dim()
```

```
dim(floral_data)
```

```
## [1] 966 8
```

```
# head()
```

```
head(floral_data)
```

```
## self_inc site latitude_N plant_id umbel_angle_deg flower_number  
## 1 SI CMGA 34.22204 CMGA.2010.1 140 18  
## 2 SI CMGA 34.22204 CMGA.2010.10 90 8  
## 3 SI CMGA 34.22204 CMGA.2010.11 120 18  
## 4 SI CMGA 34.22204 CMGA.2010.12 180 22  
## 5 SI CMGA 34.22204 CMGA.2010.13 130 21  
## 6 SI CMGA 34.22204 CMGA.2010.14 95 11  
## face_diam_mm tube_length_mm  
## 1 11.24 13.080
```

```
## 2      10.51      12.315
## 3      11.14      13.880
## 4      10.49      15.585
## 5      10.76      14.615
## 6      11.02      15.965
```

```
# tail()
```

```
tail(floral_data)
```

```
##      self_inc site latitude_N    plant_id umbel_angle_deg flower_number
## 961      SC  COSA    41.4419 COSA.2011.4          60           11
## 962      SC  COSA    41.4419 COSA.2011.5          60           8
## 963      SC  COSA    41.4419 COSA.2011.6          75          12
## 964      SC  COSA    41.4419 COSA.2011.7          75          10
## 965      SC  COSA    41.4419 COSA.2011.8          90           9
## 966      SC  COSA    41.4419 COSA.2011.9          70          11
##      face_diam_mm tube_length_mm
## 961      4.20275      5.11550
## 962      4.66175      5.49875
## 963      5.90950      5.20450
## 964      4.98600      5.60450
## 965      5.68650      5.07525
## 966      4.15600      4.42300
```

```
# str()
```

```
str(floral_data)
```

```
## 'data.frame':    966 obs. of  8 variables:
## $ self_inc      : chr  "SI" "SI" "SI" "SI" ...
## $ site          : chr  "CMGA" "CMGA" "CMGA" "CMGA" ...
## $ latitude_N    : num  34.2 34.2 34.2 34.2 ...
## $ plant_id      : chr  "CMGA.2010.1" "CMGA.2010.10" "CMGA.2010.11" "CMGA.2010.12" ...
## $ umbel_angle_deg: int  140 90 120 180 130 95 105 130 110 115 ...
## $ flower_number : int  18 8 18 22 21 11 10 16 14 12 ...
## $ face_diam_mm  : num  11.2 10.5 11.1 10.5 10.8 ...
## $ tube_length_mm: num  13.1 12.3 13.9 15.6 14.6 ...
```

3. Calculate Basic Summary Statistics

For `face_diam_mm`, calculate the following separately for SI and SC plants:

- sample size
- mean
- median
- standard deviation

Use `na.rm = TRUE` where appropriate.

```
SI <- subset(floral_data, self_inc == "SI")
SC <- subset(floral_data, self_inc == "SC")
```

```
# SI plants
summary(SI)
```

```
##      self_inc      site      latitude_N      plant_id
## Length:866      Length:866      Min.      :34.22      Length:866
## Class :character Class :character 1st Qu.:34.41      Class :character
## Mode  :character Mode  :character Median :35.30      Mode  :character
##                                     Mean  :35.64
##                                     3rd Qu.:36.95
##                                     Max.  :36.97
##
## umbel_angle_deg flower_number      face_diam_mm      tube_length_mm
## Min.      : 30.0      Min.      : 6.00      Min.      : 5.338      Min.      : 7.561
## 1st Qu.:100.0      1st Qu.:13.00      1st Qu.: 9.600      1st Qu.:12.344
## Median :110.0      Median :16.00      Median :10.697      Median :13.298
## Mean   :114.6      Mean   :15.69      Mean   :10.544      Mean   :13.369
## 3rd Qu.:125.0      3rd Qu.:18.00      3rd Qu.:11.617      3rd Qu.:14.314
## Max.   :225.0      Max.   :28.00      Max.   :15.084      Max.   :19.100
## NA's    :4
```

```
sd(SI$face_diam_mm)
```

```
## [1] 1.693516
```

```
# SC plants
summary(SC)
```

```
##      self_inc      site      latitude_N      plant_id
## Length:100      Length:100      Min.      :40.78      Length:100
## Class :character Class :character 1st Qu.:40.78      Class :character
## Mode  :character Mode  :character Median :41.36      Mode  :character
##                                     Mean  :41.22
##                                     3rd Qu.:41.44
##                                     Max.  :41.44
##
## umbel_angle_deg flower_number      face_diam_mm      tube_length_mm
## Min.      : 60.00      Min.      : 8.00      Min.      :4.156      Min.      :4.357
## 1st Qu.: 80.00      1st Qu.:11.00      1st Qu.:5.294      1st Qu.:5.896
## Median : 90.00      Median :13.00      Median :5.974      Median :6.176
## Mean   : 88.28      Mean   :13.27      Mean   :5.960      Mean   :6.213
## 3rd Qu.: 95.00      3rd Qu.:15.00      3rd Qu.:6.655      3rd Qu.:6.617
## Max.   :130.00      Max.   :25.00      Max.   :7.935      Max.   :7.770
## NA's    :1
```

```
sd(SC$face_diam_mm)
```

```
## [1] 0.8930541
```

Part 2 — Visualize SI and SC Populations

Load ggplot2.

```
library(ggplot2)
```

4. Faceted Histogram

Create a histogram of `face_diam_mm`.

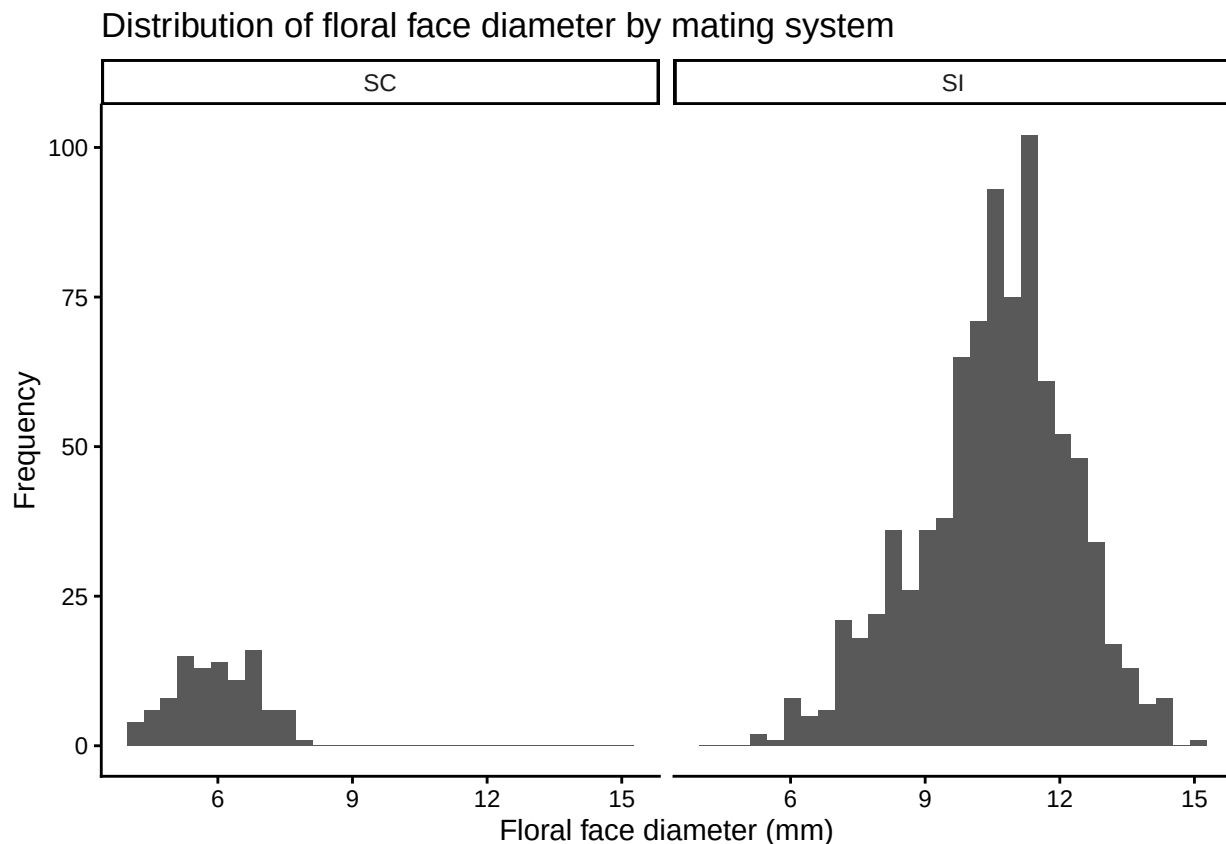
Your plot should:

- show separate panels for SI and SC plants using `facet_wrap()`
- use an appropriate `binwidth`
- use `theme_classic()`
- include appropriate axis labels with units
- include a descriptive title

Create your histogram here.

```
ggplot(floral_data, aes(x= face_diam_mm)) + geom_histogram() + facet_wrap(~ self_inc) + theme_classic()
  x = "Floral face diameter (mm)",
  y = "Frequency",
  title = "Distribution of floral face diameter by mating system"
)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



5. Figure for Umbel Angle by Mating System

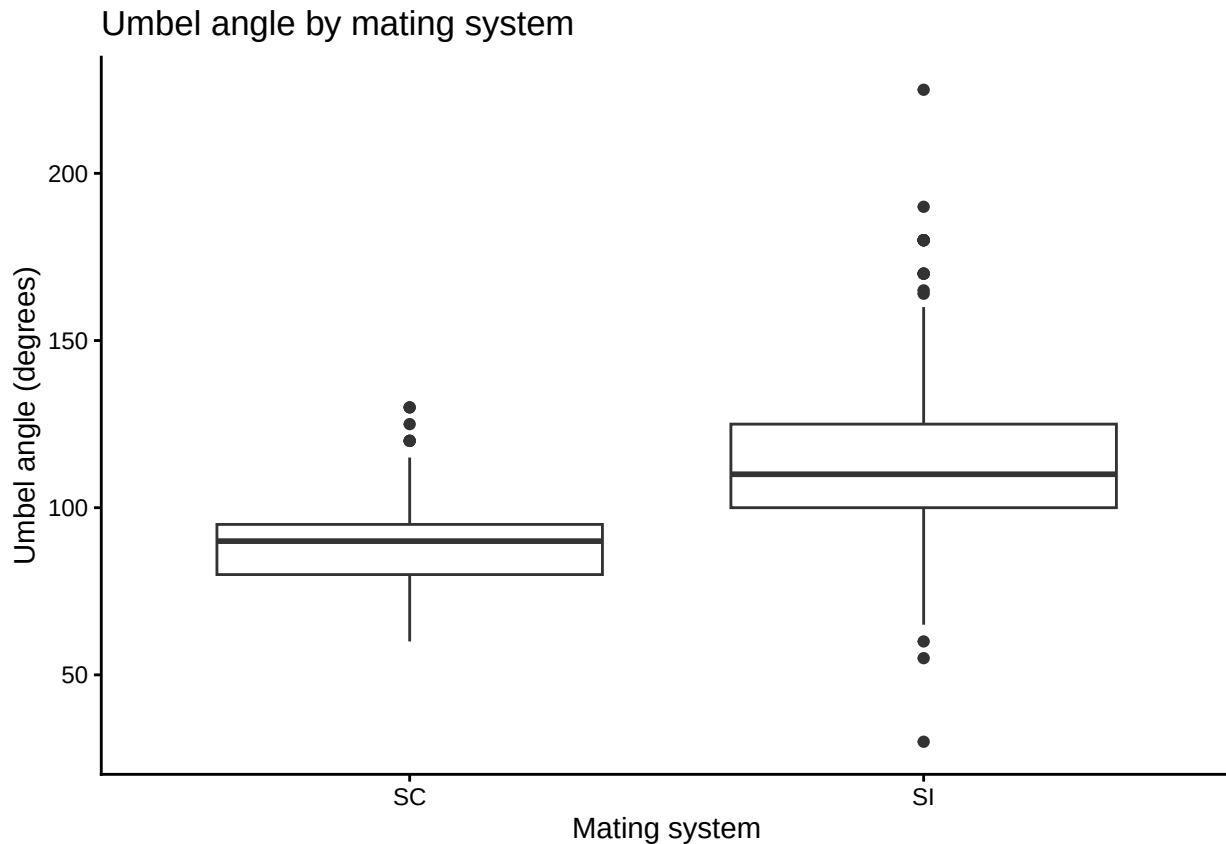
Create a appropriate figure comparing `umbel_angle_deg` between SI and SC plants.

Your plot should:

- use `facet_wrap()` if needed
- use `theme_classic()` for theme
- include axis labels, appropriate title, and captions
- only the Figure should be present in the knitted document

```
ggplot(floral_data,aes(x = self_inc,y = umbel_angle_deg)) +
  geom_boxplot() +
  theme_classic() +
  labs(x = "Mating system",y = "Umbel angle (degrees)",title = "Umbel angle by mating system")
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Part 3 — Relationship Between Floral Traits

6. Floral Face Diameter vs. Tube Length

Create a figure showing the relationship between:

- face_diam_mm
- tube_length_mm

Your plot should:

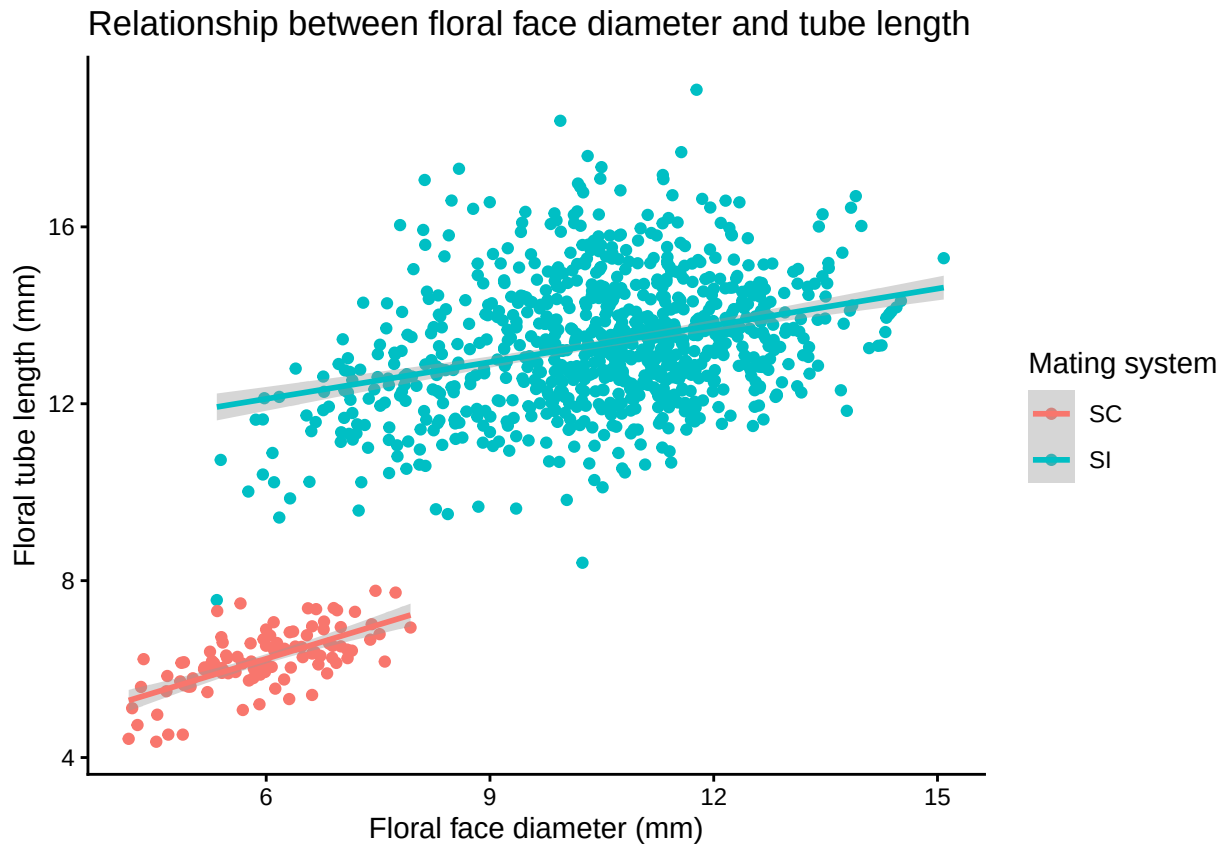
- colour points by `self_inc`
- add a separate linear regression line for each mating system with standard error
- use `theme_classic()`
- include axis labels with units, appropriate title, legends and captions

Create your figure here.

```
ggplot(floral_data,
  aes(x = face_diam_mm,y = tube_length_mm,colour = self_inc)) +
```

```
geom_point() +
geom_smooth(method = "lm",se = TRUE) +
theme_classic() +
labs(x = "Floral face diameter (mm)",y = "Floral tube length (mm)",colour = "Mating system",
title = "Relationship between floral face diameter and tube length")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



7. Interpretation

Explain your interpretation based on the figures generated above.

Student should say SC plants have less investment on floral traits which has association to pollination attraction.